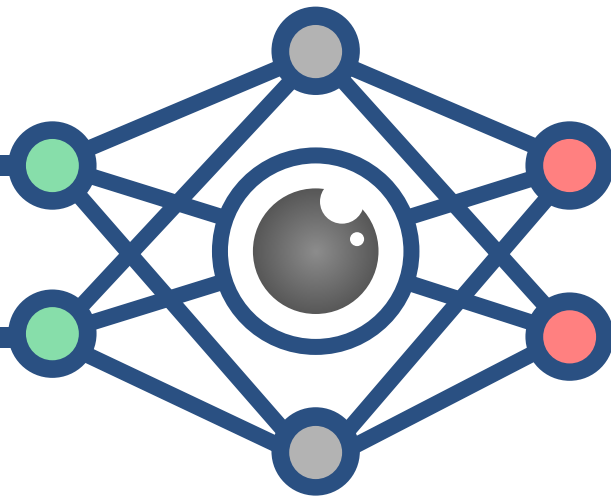


CS3485

Deep Learning for Computer Vision



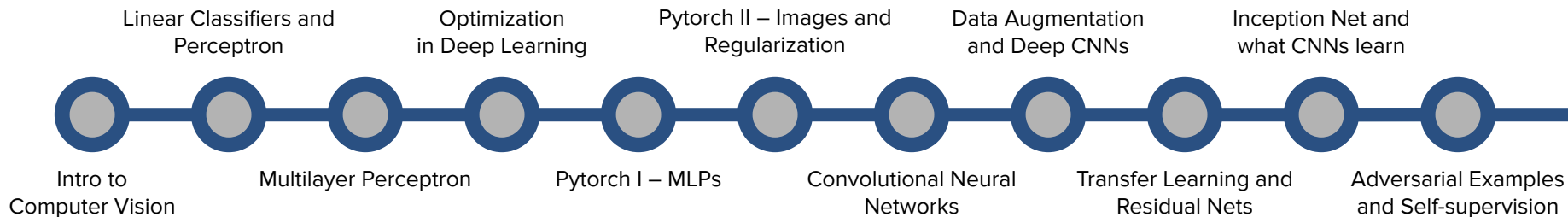
Lec 12: The Physical Cost of Deep Learning

Announcements

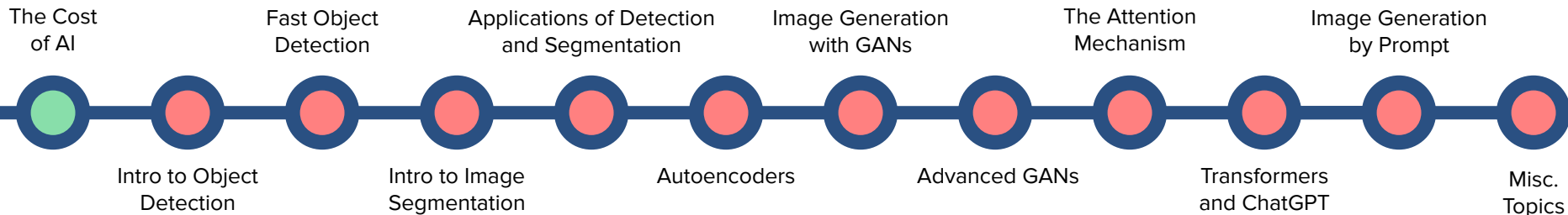
- We'll have the quiz at the beginning of the lecture today!

(Tentative) Lecture Roadmap

Basics of Deep Learning



Deep Learning and Computer Vision in Practice



Today's Lecture

- Welcome to my first “Contested Values in Computer Vision: Collective Voices, Constructive Viewpoint, Conversations and Videos” (CVCVCVCVCV or CV5)
- So far, training a network has been a few lines of code for us: `model.to(device)` and a loop.
- Today, we look at **where that code actually runs**, what it **costs**, and why it has become one of the most debated topics of 2026.
- I also hope to have some discussion! This is the plan:



Recap: Transfer learning and ResNets

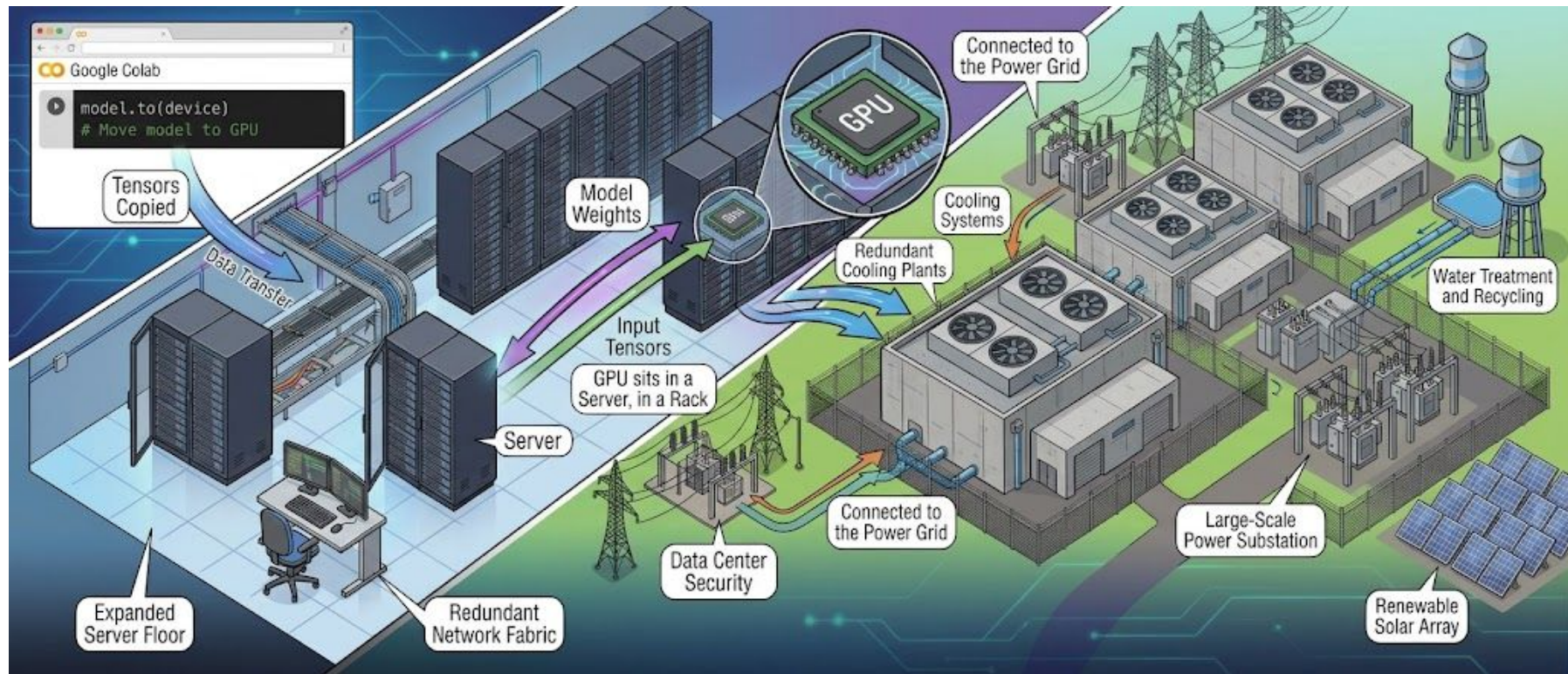
- Last time, we adapted a **pretrained** VGG16 to Cats vs. Dogs: we froze its feature extractor and trained only **~65k parameters**, which took a few minutes on Colab.
- That was cheap only because someone had already paid for the expensive part: training the network on the **1.28 million** images of ImageNet, for dozens of epochs.

```
model = models.resnet50(weights=models.ResNet50_Weights.IMAGENET1K_V1)
model = model.to(device)
```

- When we call `model.to(device)` on Colab, our tensors are copied to a **GPU** that sits in a server, in a rack, in a building connected to the power grid and to a water supply.
- That building is a **data center**. Today, we'll look inside one, and then at why so many people are talking about them.



The Anatomy of a Data Center



How much compute does training take?

- Recall that a ConvLayer is just a (huge) number of **multiply-adds**. ResNet50 needs about **7.6 billion floating-point operations (FLOPs)** to classify a single 224×224 image.
- Training costs roughly $3\times$ a forward pass (the backward pass costs about twice the forward one). So, for **90** epochs on ImageNet:

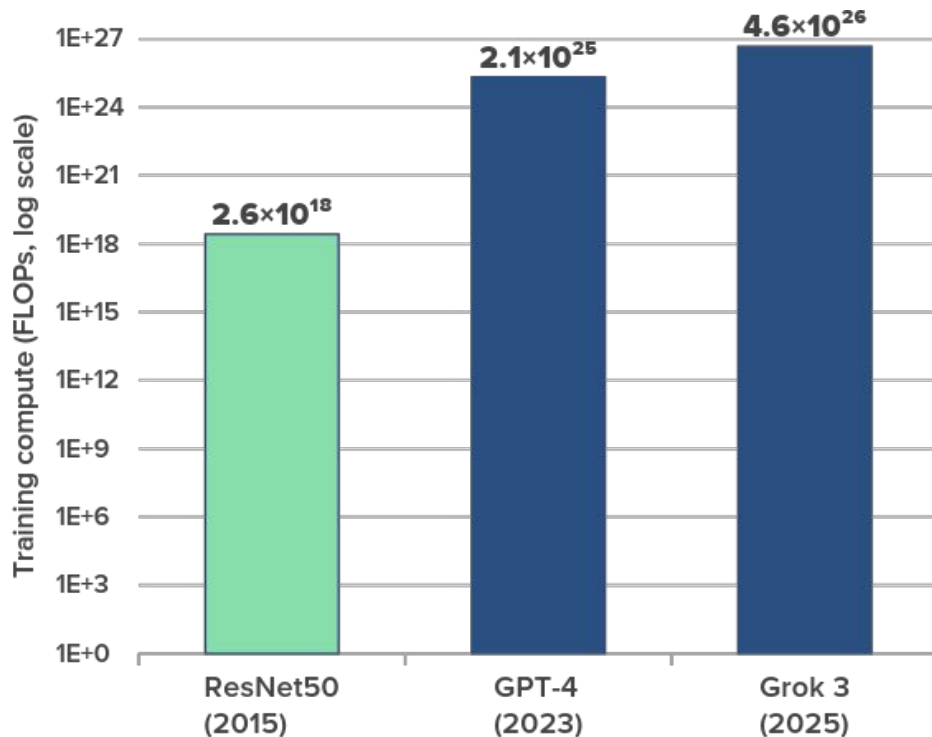
```
flops_fwd = 7.6e9           # ResNet50, one 224x224 image (forward pass)
n_images, n_epochs = 1.28e6, 90  # ImageNet training set
train_flops = 3 * flops_fwd * n_images * n_epochs
print(f"{train_flops:.1e} FLOPs")
```

2.6e+18 FLOPs

- That sounds like a lot, but one modern GPU (an NVIDIA **H100**) sustaining about 4×10^{14} **FLOP/s** would finish it in **under 2 hours**, drawing ~ 700 **W**: about **1.3 kWh** in total, less than a space heater running for an hour.

From ResNet50 to frontier models

- Today's largest models are trained with **10 to 100 million times** more compute than our ResNet50
- And the trend is steep: Epoch AI estimates that the training compute of frontier models has grown by **4 – 5× per year**.
- A single GPU is no longer enough. Grok 3, for example, was trained on xAI's **Colossus** cluster in Memphis, with about **100,000 H100 GPUs**.



* [Source](#): “Training compute of frontier AI models grows by 4-5x per year”, 2024

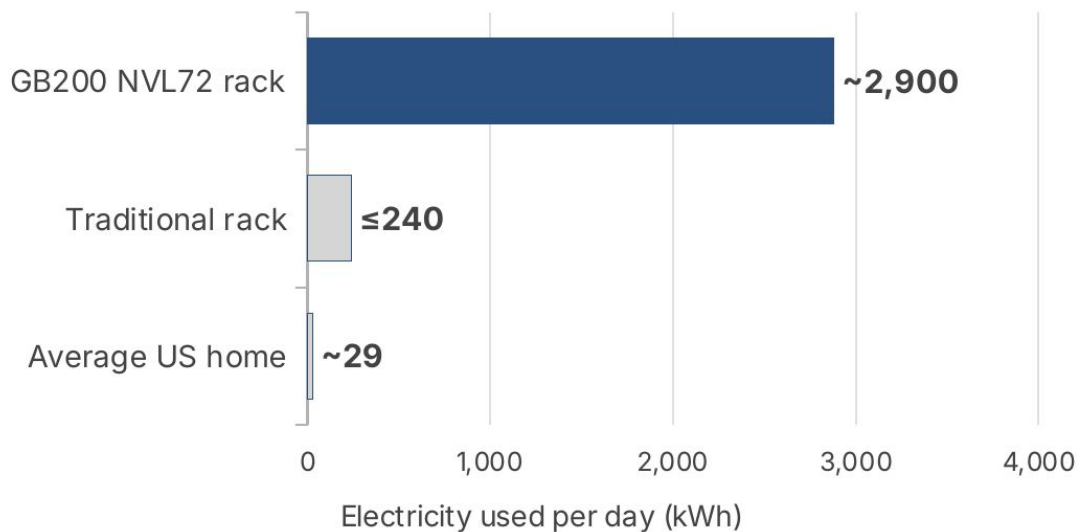
What's inside a data center?

- A data center is a building with three jobs:
 - **Deliver power** to the servers,
 - **Run the computation**,
 - **Remove the heat** it produces.
- **Cooling:** heat is carried away by air or liquid loops and dumped outside, often by evaporating water in cooling towers.
- **Key fact:** nearly every watt that enters a GPU leaves it as heat. The more we compute, the more we have to cool.



AI racks are different

- A traditional server rack draws about **3–10 kW**. An NVIDIA **GB200 NVL72** rack (72 GPUs) draws about **120 kW**: an order of magnitude more, in the same floor space.



- One AI rack draws as much power, on average, as about **100 US homes**.
- At this density, air alone can't remove the heat: these racks require **direct-to-chip liquid cooling**, with cold plates on each chip connected to the facility's water loop.

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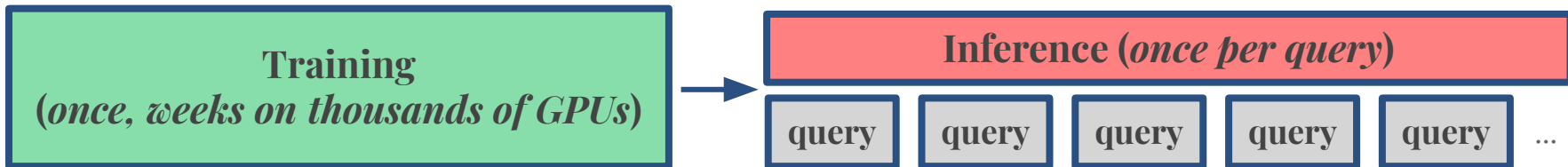
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Training vs. inference

- So far we talked about **training**: a huge job done once. When a trained model is *used* (a chatbot answer, for example) we run **inference**, which happens **billions of times a day**.

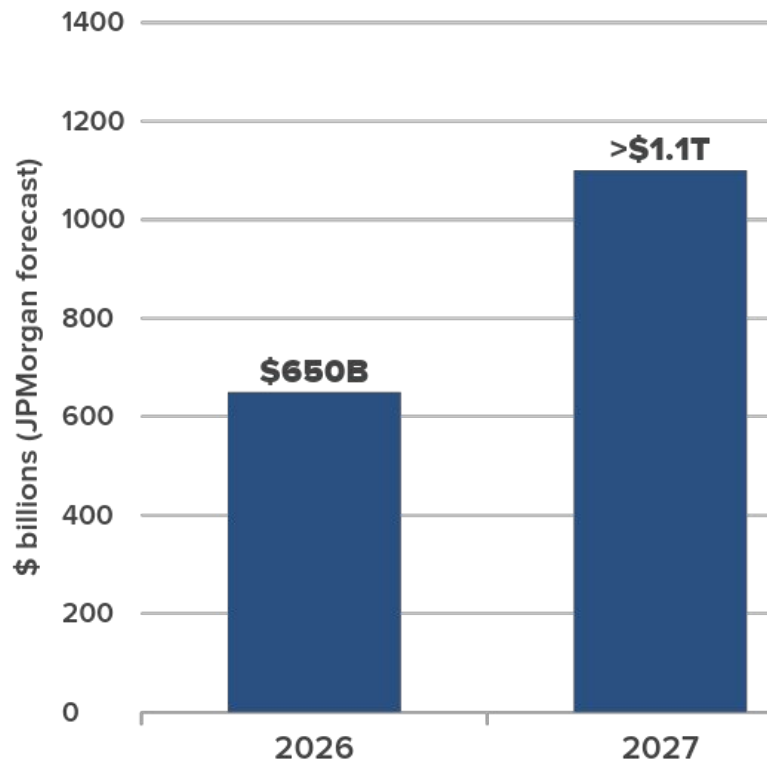


- For widely used models, the energy spent on inference over the model's lifetime can **exceed** the energy spent training it. For ChatGPT*, for example:
 - **Query volume**: about 2.5 billion prompts per day.
 - **Energy per query**: about 0.34 Wh on average,
 - **Total per day**: $2.5 \times 10^9 \times 0.34 \text{ Wh} \approx 850 \text{ MWh}$.
 - **Over a year**: $\approx 310 \text{ GWh}$. That's an average draw of about 35 MW, around the clock, or about what Portland (ME) consumes in a year.

* Sources: "[The Gentle Singularity](#)", 2025; "[Altman plans D.C. push to 'democratize' AI economic benefits](#)", 2025

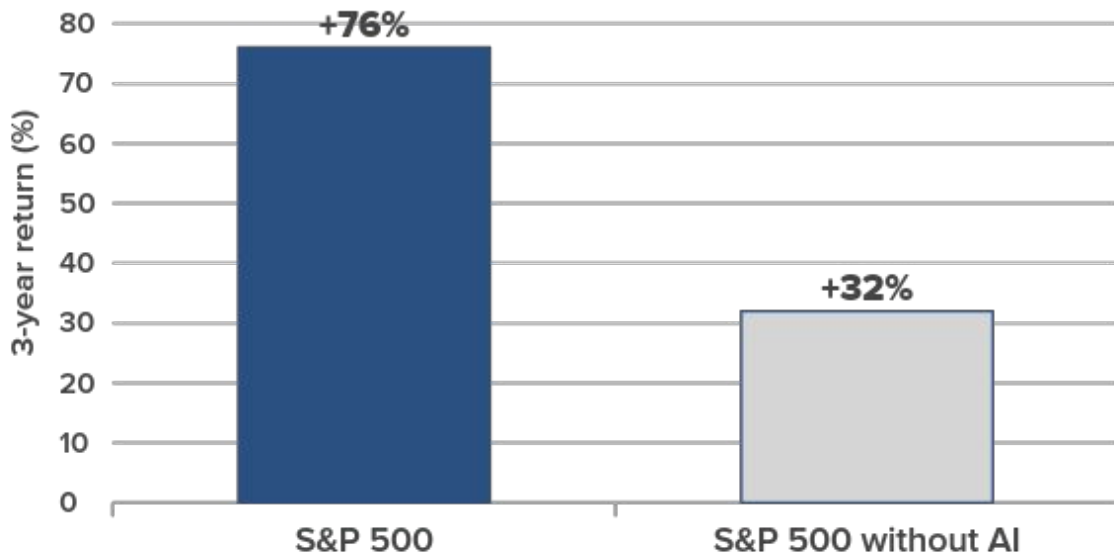
The Economics of Data Centers

- Why are so many data centers being built so fast? Because the largest tech companies are investing at an unprecedented scale.
- JPMorgan expects the **hyperscalers'** (Microsoft, Amazon, Alphabet, Meta, Oracle...) capital expenditures to reach **\$650 billion** in 2026 and to exceed **\$1.1 trillion** in 2027.
- For scale: Big Tech's 2026 AI spending alone already rivals the size of **Sweden's entire economy** (Fortune).
- JPMorgan estimates total AI-related capex of **\$5.5 trillion** through 2030, increasingly financed with **debt**.



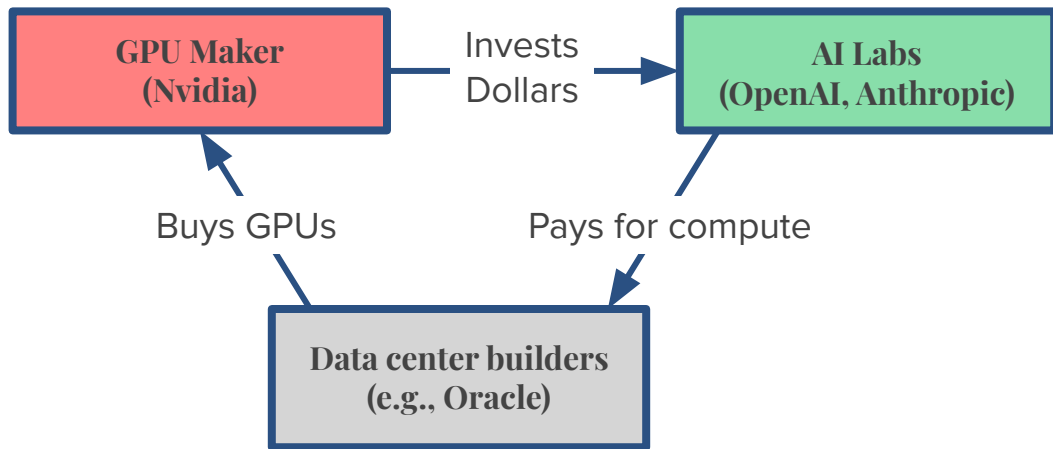
The stock market is betting on it

- According to Goldman Sachs, AI-related companies make up about **45%** of the S&P 500.
- Over the three years to February 2026, the index rose **76%**; without its AI companies, it rose only **32%**.
- In other words: if you own an index fund (say, in a retirement account), you own a **large bet on the data center build-out**, whether you know it or not.



Circular deals

- Many deals in the AI industry are **circular**: a supplier invests in its customer, who then uses that money to buy the supplier's products.



- **Why it matters:** if AI revenue falls short, the losses hit every link of the chain at once.

- **Examples:**

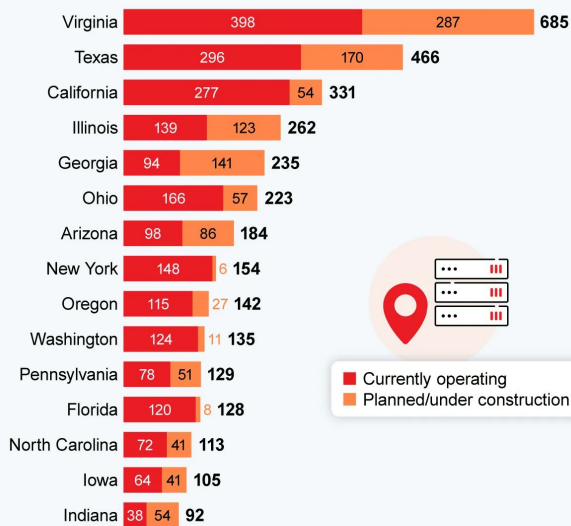
- Sep. 2025: Nvidia announces plans to invest up to **\$100B** in OpenAI; the talks stalled in early 2026.
- OpenAI signs a **\$300B** cloud contract with Oracle.
- Jul. 2026: Nvidia reportedly discusses a **\$250B** backstop for a 10 GW OpenAI campus in Ohio.

And for the towns where they are built?

- Data centers bring **tax revenue** and **construction jobs**, which is why states compete for them with **tax breaks**.
- According to Business Insider, **80%** of data center jobs are in **construction**: once a it is built, few people are needed to run it.
- And who pays for the bill is **disputed**:
 - Evidence suggests the costs are already being **passed on to customers**.
 - Others suggest that **no statistically significant relationship** between data center concentration and faster electricity rate increases.
- These are exactly the points on which our two videos disagree. Let's watch them.

Which U.S. States Have The Most Data Centers?

Number of data centers by U.S. state as of February 2026



Source: Pew Research Center



Video 1: Business Insider

- **Exposing The Dark Side of America's AI Data Center Explosion** (Business Insider, Sept. 2025, ~30 min). An investigative piece, nominated for a 2026 News & Documentary Emmy.
- While you watch, write down:
 - a. One claim you could **check with data**.
 - b. One number that **surprised** you (compare it with your warm-up guesses!).
 - c. One thing the video **doesn't** tell you.



Video 2: John Stossel

- **Data Center Freak-Out** (Stossel TV, May 2026). An **opinion** piece by a libertarian commentator: it argues that the backlash is overblown and that government rules are the real problem.
- While you watch, separate:
 - a. **Evidence:** claims backed by a number or a source you could check.
 - b. **Framing:** word choices meant to persuade .
 - c. Where does it **agree** with Business Insider?



Discussion (*In pairs or groups*)

- Summaries:
 - What is each video about? What are the main claims in each video?
 - How do they back up their claim? Think of where the argument is coming from: emotional, social, economical, etc.
 - What is a positive and a negative aspect of each video in terms of narrative/argument?
- One vs the other:
 - Which claim held up best? Why?
 - Which framing was the most misleading?
- Overall morals:
 - What are the major issues involved in Data Centers and their constructions?
 - Why is this topic controversial?
 - Isn't there a simple way out of this?
- Finally:
 - What is your overall opinion of data centers?